Trenitalia
ETR1000 VZI-6

Aux Electrical Supply

Revision	Author	Date	Revision Description
00.1	G. De Marco	29/08/2023	Draft Release Platform-.Zefiro-Europe/2F_06.01.-.Aux Electrical Supply-VF.-.Zefiro-Europe/Aux Electrical Supply-VF.CONCEPT, baseline 2, doc 3EST000225-7914_B reviewed for VZI-6
00	G. De Marco	18/10/2023	Issue for VZI-6 PDR
1.01	G. De Marco	16/04/2024	Added requirement HR01, HR02, HR03, HR04, HR05, HR06
01	G. De Marco	30/04/2024	Issue for BL1
2.01	G. De Marco	08/11/2024	Mod. req 2F_06.01.Zefiro-Europe.CONCEPT.322 (reset via msg data)
2.02	G. De Marco	08/01/2025	Added requirement HR07
02	G. De Marco	24/01/2025	Issue for functional BL3
03	G. De Marco	23/07/2025	Issue for functional BL4: - Added requirement HR08 - Modified requirement 322

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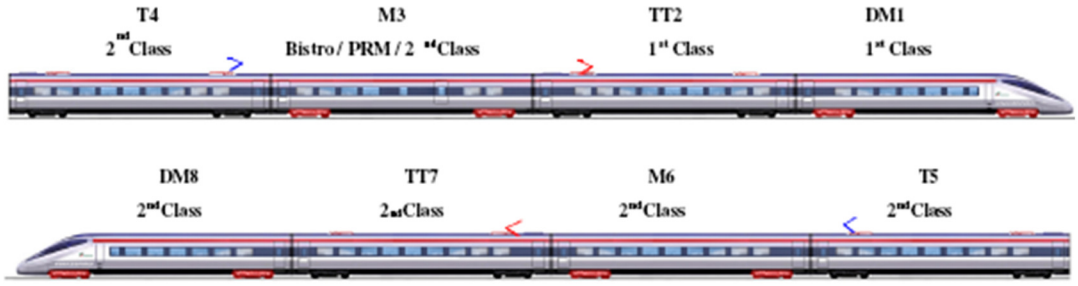
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1 Introduction

This document is a function specification for the control of the a*uxiliary electric* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
I/O	Input / Output unit
TCMS	Train Control and Management System
RCD	Residual Current Protection Device
ACM	Auxiliary Converter Module
CCM	Chopper Converter Module
LCM	Line Converter Module
MCM	Motor Converter Module
PCU	Propulsion Control Unit
TT-system	Type of network configuration, directly earthed

2 System Overview

2.1 System Configuration

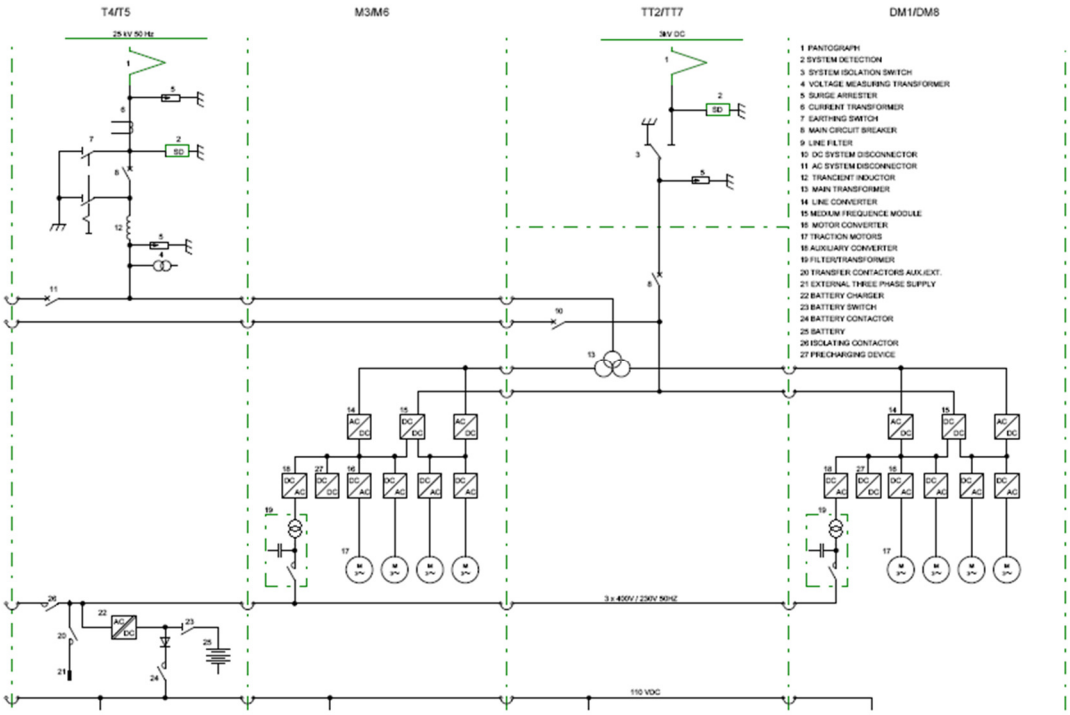
The auxiliary electric system is designed to supply power to different loads in the 8-car train.

- Three-phase 230/400 VAC, 50 Hz
- 110 VDC

Coupling between these systems is performed by 3AC-bus supplied Battery charger.

The 3AC 400V- bus is designed as short circuit proof cable and laying. All rotating devices in the train require a clockwise rotating field. This has to be made sure for external supply.
The installed electrical system is the TT-System. All metallic in normal operation touchable surfaces and housings of the electrical equipment are connected to carbody with potential equalisation. The carbody is connected to earth (rail) with brushes at undriven axles and the connected PE is separately wired to the neutral wire. The neutral wire is connected to the star-point of the transformers of the ACMs. This connection shall be disconnected if the ACM-contactor is not closed and will be supervised by TCMS.

In cases of external supply the TT-system will be retained. The external power supply provides the supply with 5 wire cables exclusive pilot wires.
RCDs are required for auxiliary consumers without permanent connection.
An overview of the main circuits of electrical system of the V300ZEFIRO is shown in figure below.



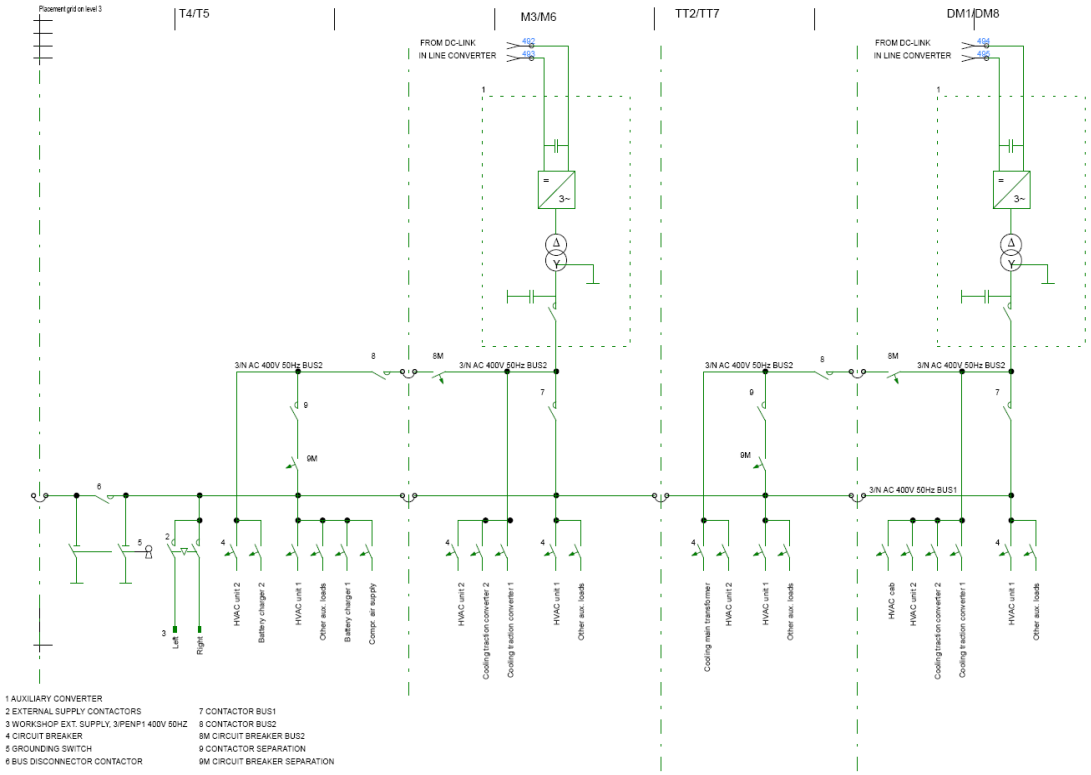
Main Circuit Diagram

3 Functional requirements

3.1 Electrical Auxiliary System (3~ 400V AC)

3.1.1 Function General

The train AC network is composed of two 4-car units, connected with a bus disconnector. In each 4-car unit of the train there are 2 ACM and these ACM are operating synchronized in parallel. In normal mode the network disconnector is closed and 4 ACM are operating synchronized in parallel.
An overview of circuit diagram of Three-phase Auxiliary System is shown in figure below



Three-phase Auxiliary System

A single failure in the auxiliary systems must not cause the loss of tractive effort and electric braking effort.

3.1.1.1 Function Design

During the start up of the ACMs (all 4) the auxiliary loads have to be activated with time delays to minimize the current rise. The battery charger will start as soon as 3-phase is available on the bus among with traction fans, water and oil pumps etc. To supervise the state and guarantee the running of the power supply the TCMS system controls the main contactors and disconnectors that are necessary for the load management and the safety and comfort functions of the train.
When at least 2 ACM:s are started the auxiliary loads are activated with time delays. Activation of the loads is depending on how many ACM:s are active.

3.1.1.2 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.120
Description	The neutral point in auxiliary three-phase shall be directly connected to the carbody.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.121
Description	It shall be possible to disconnect the neutral conductor from the carbody.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.152	The ACM shall have 4 pole feed contactor.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.191	The neutral of the 3-phase bus shall be connected to ground via ACM load contactor.	Derived Requirement	4S_05.-.Propulsion		N/A

ID	2F_06.01.Zefiro-Europe.TRS.119
Description	The auxiliary loads shall be activated in sequence with time delays to minimize the peak load current. This applies to all operational scenarios.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.133
Description	A power management function shall ensure to take care of the ratio between available power and the load of consumers if power limitations apply. This is the case for external 3~ power supply and any degraded mode where more than one ACM are faulty.
Safety level	N/A

Smaller loads like traction fans, water and oil pumps can start when 3 phase is available on the 3 phase bus. Larger loads like HVAC and Main Air Compressor must be started in a sequence to limit the start current.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.187	When at least two ACM's have been producing 3 phase for 2 seconds or more the 3 phase load startup sequence shall begin.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.110	The loads shall be started in the following order when at least two ACMs have started: – Main Air Compressor 1 – Main Air Compressor 2 – HVAC in active cab – HVAC 1 in DM1 – HVAC 1 in TT2 – HVAC 1 in M3 – HVAC 1 in T4 – HVAC 1 in T5 – HVAC 1 in M6 – HVAC 1 in TT7 – HVAC 1 in DM8 – HVAC 2 in DM1 – HVAC 2 in TT2 – HVAC 2 in M3 – HVAC 2 in T4 – HVAC 2 in T5 – HVAC 2 in M6 – HVAC 2 in TT7 – HVAC 2 in DM8 – HVAC in DM1 Cab (if cab not activated) – HVAC in DM2 Cab (if cab not activated)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.321	When feedback have been received that a load has started there shall be a 2 second delay until the next load in the priority order is allowed to start.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.322	If feedback, that an HVAC load has started, has not been received within 20 seconds, start enable command to the HVAC compressor will be removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_06.01.Zefiro-Europe.CONCEPT.275	Battery charger shall start when 3 phase is available on the 3 phase bus.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.HR01	Main transformer fans, Traction fans, oil pump, water pump shall start when 3 phase is available on the 3 phase bus.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.188	When there is a need for the traction fans, water and oil pumps to run (according to Propulsion control) and 3 phase is available on the 3 phase bus the load shall start.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.487	Traction fans, water and oil pumps shall start when 3 phase is available on the 3 phase bus.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.2 Activation and Synchronisation of ACM

3.1.2.1 Function Design

When the 3AC bus is not energised by external power supply and line voltage is available the ACM starts. Only one ACM is allowed to start with an idle 3AC-bus and close its load contactor. After the first ACM supplies the 3AC-bus the other ACMs starts successive by synchronizing before connecting. Dependent on the amount of running ACMs the 3AC loads will be connected to the 3AC-bus.

3.1.2.2 Normal operation

First ACM is started on an idle bus and the others are synchronized one at a time.

3.1.2.3 Failure cases

1. ACM fails to start

3.1.2.4 Failure detection

1. If ACM is not started an event is stored in ACM itself

3.1.2.5 Failure handling

1. If an ACM fails to start the HVAC will be reduced.

3.1.2.6 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.107
Description	When high voltage (on DC-link) is detected, the ACM's shall be started. One ACM is to be defined to start first, the other ACM´s shall start with synchronising their 3~AC output voltage.
Safety level	N/A

Four ACM modules are feeding into one common 400V AC bus per 8 car train. The ACM modules need to synchronise to work in parallel.
When high voltage (on DC-link) is detected, the ACM's shall be started. One ACM is to be defined to start first, the other ACM´s shall start with synchronising their 3~AC output voltage.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.444	Only one ACM shall be allowed to start at the time and there shall be a 500ms delay time between each start attempt.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.447	An ACM shall be allowed to start if the following conditions are fulfilled • ACM is not faulty (signal "ACM faulty"=False from ACM) • ACM is not cut out • Communication to ACM is ok	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	• For the DC-link where ACM is connected: Signal "DC-link charged" from PCU is set.				
2F_06.01.Zefiro-Europe.CONCEPT.HR02	If all DC-link charged are set, the starting sequence for ACM will be: • DM1 • M3 • M6 • DM8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.553	None ACM may be allowed to start, if an external power supply is recognised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.446	If the 3-phase bus is inactive (signal "ACM bus active"=False from ACM) the ACM shall be ordered to do an idle start by signal command: “disable idle start of auxiliary converter” = False and signal command: “disable start of auxiliary converter” = False until the ACM has started (signal "ACM in operation" = True) or for maximum 10 seconds.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.445	If the 3-phase bus is active (signal ACM bus active=True from ACM) the ACM shall be ordered to do a synchronized start by signal command: “disable idle start of auxiliary converter” = True and signal command: “disable start of auxiliary converter” = False until the ACM has started (signal "ACM in operation" = True) or for maximum 10 seconds.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.57	When an ACM gets an order to do a "idle start" by signals "Command: disable idle start of auxiliary converter" = False and "Command: disable start of auxiliary converter" = False from TCMS and DC voltage is detected the ACM shall start not using phase rotation detection as the 3*400V bus is powerless.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.284	When an ACM gets an order to do a "synchronized start" by signals "Command: disable idle start of auxiliary converter" = True and "Command: disable start of auxiliary converter" = False from TCMS and DC voltage is detected the ACM shall start using phase rotation detection as the 3*400V bus is powered.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.569	whenever an ACM starts the TCMS has to rise an event stating that the specific AUX converter ha started and the starting mode (syncronized/not syncronized)	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.3 ACM reset

3.1.3.1 Function Design

The ACM normally protect it self by disconnecting from the 3 phase bus if any abnormality is detected. To prevent the ACM from starting and stopping when it should not from an over all perspective the train control sends enable and reset signals to control the operation of the ACM

3.1.3.2 Normal operation

When enable signal is given the ACM shall start. If the ACM does not start when enabled to start another ACM will be enabled to start.

3.1.3.3 Failure cases

1. Enable signal is given but ACM is not started
2. Reset signal is given but ACM is not reset

3.1.3.4 Failure detection

1. If ACM is not started it’s considered as failed. Event is stored in ACM itself

2. If ACM is not reset it's considered as failed. Event is stored in ACM itself

3.1.3.5 Failure handling

1,2. If one ACM fails to start the loads needs to be reduced.
Two ACMs fails to start. Instant workshop

3.1.3.6 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.154
Description	In case of fault on the 3 phase bus, the ACM shall automatically isolate. It shall be permanently isolated until a reset is recived from the train control.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.108
Description	Each ACM shall be equipped with a load contactor. This contactor shall be able to disconnect the converter if a failure occurs.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.62	If the ACM detects any fault, within its own fault detection system, that requires disconnection from the 3 phase bus the ACM shall disconnect it self using the ACM load contactor controlled by ACM.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.246	If the ACM detects any fault, within its own fault detection system, that requires disconnection from the 3 phase bus and has reached the maximum restart attemps, the ACM shall stay disconnected from the 3 phase bus until a reset is received from train control.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.315	When master reset is performed the Reset signal shall be set to ACM.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.4 AC on Board Supply System

3.1.4.1 Normal operation

The load contactors 7, 8 and 6 are closed in normal operation (see figure shown at par. 3.1.1). These contactors are opened at fault conditions. When all ACM:s are operating there are no restrictions for the loads.

3.1.4.2 Failure cases

1. System indicates fewer ACMs running that actually are providing power to the auxiliary bus.
2. System indicates more ACMs running that actually are providing power to the auxiliary bus.
3. Contactor (7) does not close/open on order
4. Contactor (8) does not close/open on order
5. Contactor (9) does not close/open on order
6. Contactor (6) does not close/open on order
7. Short circuit in distribution box, same car as ACM is located
8. Short circuit in distribution box T- or TT car
9. Short circuit on the bus between M- and T car
10. Short circuit main bus

3.1.4.3 Failure detection

1. Can not be supervised. HVAC will be reduced in the same way as if ACM is out of order.
2. Overload of the ACMs is supervised by the ACM itself. Event is stored in ACM itself
3. When the load contactor(7) is not possible to operate an event will be set
4. When the load contactor(8) is not possible to operate an event will be set

5. When the load contactor(9) is not possible to operate an event will be set
6. When the load contactor(6) is not possible to operate an event will be set
7. All ACM:s detects short circuit
8. All ACM:s detects short circuit
9. All ACM:s detects short circuit
10. All ACM:s detects short circuit

3.1.4.4 Failure handling

1. HVAC needs to run in degraded mode.
2. HVAC needs to run in degraded mode.
3. If contactor (7) is not possible close all loads still have supply.
4. Loads supplied via contactor (9).
5. Loads supplied via contacor (8).
6. All loads will be supplied in both cases.
7. I the short circuit is in a M car then this car, distribution box will be isolated by opening contactors 7 and 8. The loads in a T/TT car will be supplied via contactor 9. Impact will be loss of 25% traction, 12.5% HVAC saloon and the cab HVAC.
8. If the short circuit is in distribution box T/TT-car, contactors 8 and 9 will be opened. If the T car is isolated the impact will be loss of 1 battery charger and 1 HVAC unit. Impact on a TT car will be loss of the transformer cooling and 1 HVAC unit. Loss of 50% traction.
9. If the short circuit is between M- and T car contactor 8 will be opened and contactor 9 closed. No impact.
10. If the short circuit is on the main bus the contactors 6 and contactors 7 on the faulty bus will be opened. No impact on traction, loss of 25% HVAC

3.1.4.5 Normal Mode

3.1.4.5.1 Function Design

When there are no short-circuits or other faults detected the auxiliary power system will start in normal mode with closed contactors to allow maximum of loads to be powered. HVAC is able to run at full power.

3.1.4.5.2 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.111
Description	When the vehicle is connected to the catenary system and all ACMs are active and supplying all three-phase loads, the auxiliary electric system is in mode “Normal operation”.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.323	When the train starts up the contactors shall be ordered as following: – Close all bus 1 and 2 separation contactor (9) for 5 seconds and then Open all bus 1 and 2 separation contactor (9) – Close all "Bus 1 contactor" (7) – Close all "Bus 2 contactor" (8) Note: When the external power supply is active, the test of contactor (9) shall not be performed in order to avoid temporary supply to the loads	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.552	The "train half separation contactor" (6) shall be ordered to close when a command to raise a pantograph has been given and there is no voltage on the 3-phase bus.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.467	The load contactor(8) shall be closed when ordered	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

2F_06.01.Zefiro-Europe.CONCEPT.468	To close the load contactor (8) the close order shall be set and to open the close order shall be reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.469	When the load contactor(8) does not follow the order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.503	If the load contactor (8) does not close on order then the load contactor (8) shall be ordered to open and the load contactor (9) on the same 3 phase bus shall be ordered to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.470	The load contactor(7) shall be closed when ordered	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.471	To close the load contactor (7) the close order shall be set and to open the close order shall be reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.472	When the load contactor(7) does not follow the order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.556	If the load contactor (7) does not close on order then the load contactor (7) shall be ordered to open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.473	The load contactor(9) shall be closed when ordered	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.474	To close the load contactor (9) the close order shall be set and to open the close order shall be reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.475	When the load contactor(9) does not follow the order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.557	If the load contactor (9) does not close on order then the load contactor (9) shall be ordered to open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.476	The load contactor(6) shall be closed when ordered	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.477	To close the load contactor (6) the close order shall be set and to open the close order shall be reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.478	When the load contactor(6) does not follow the order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.558	If the load contactor (6) does not close on order then the load contactor (6) shall be ordered to open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.HR03	If shall be possible to open the load contactor (6) from TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_06.01.Zefiro-Europe.CONCEPT.HR07	Opening command to the load contactor (6) from TOD will be allowed when 3 phase is not available on the 3 phase bus	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.HR04	An event shall be set when load contactor (6) has opened	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_06.01.Zefiro-Europe.CONCEPT.561	the status of a MV contactor shall be evaluated as stated here below: order / open feedback / closed feedback / final status / final order to be delivered close / 0 / 1 / closed / close close / 1 / 1 / closed / close close / 1 / 0 / **does not close on order** / **open** close / 0 / 0 / **does not close on order** / **open** open / 0 / 1 / **does not open on order** / **open** open / 1 / 1 / opened / open open / 1 / 0 / opened / open open / 0 / 0 / **does not open on order** / **open**	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.4.6 Degraded Mode

3.1.4.6.1 Function Design

If one or more ACM:s is out of order the HVAC needs to be reduced.

3.1.4.6.2 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.115
Description	If more than 1 ACM fails a suitable number of loads shall be automatically disconnected to compensate the loss of power supply.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.112
Description	If one ACM fails, operation shall be continued with 3 of 4 ACMs supplying all loads. Redundancy for one ACM failure shall be provided, the remaining modules shall compensate the load if possible.
Safety level	N/A

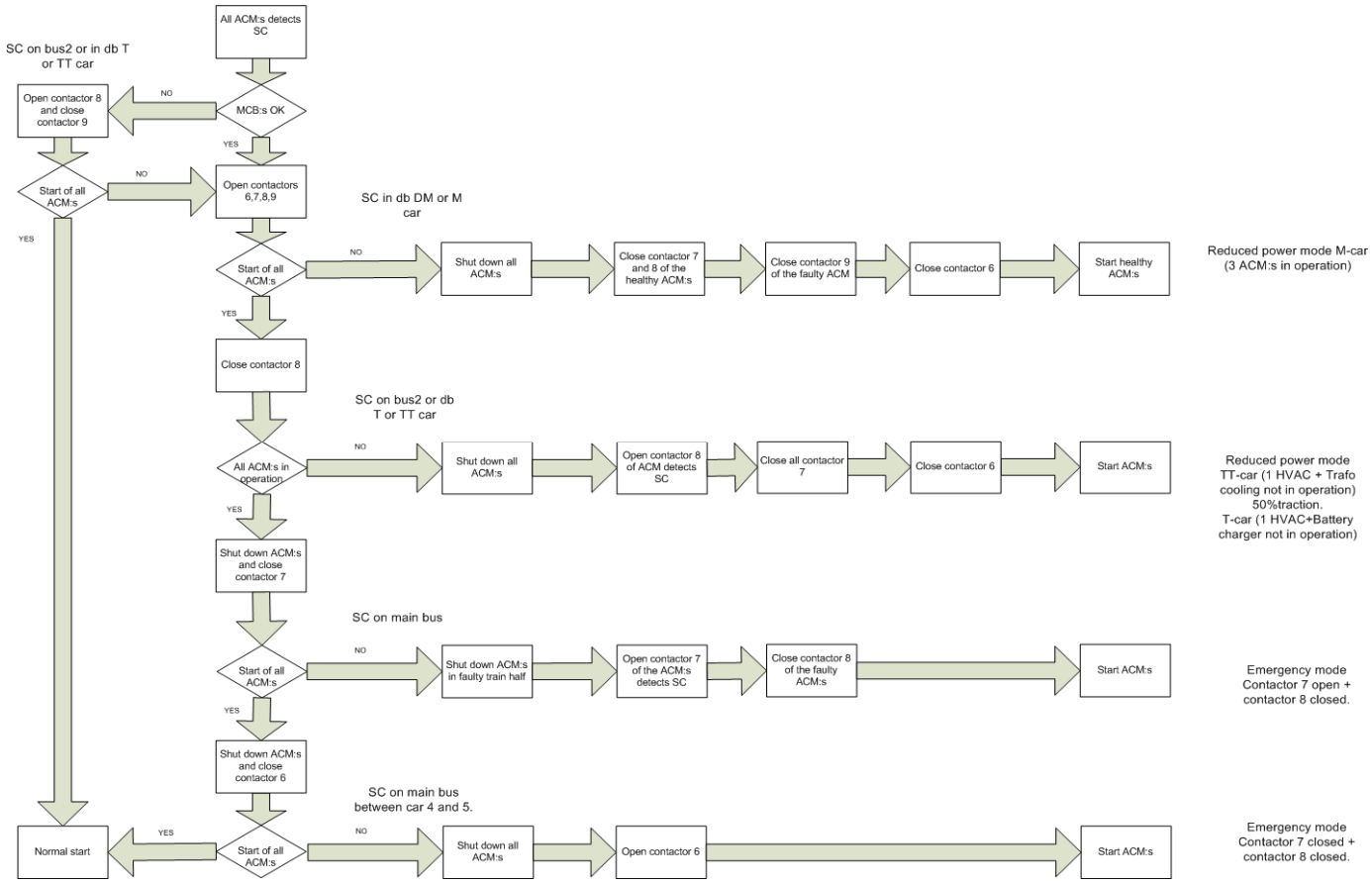
Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.440	At standstill operation on 3kV DC network (with four ACMs in operation) the HVAC maximum allowed power consumption shall be 45kVA/car. With the same conditions under 1,5kV DC network, the the HVAC maximum allowed power consumption shall be 32kVA/car.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.562	At running operation on 3kV/1,5kV DC network (with four ACMs in operation) the HVAC maximum allowed power consumption shall be 80kVA/car.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.01.Zefiro-Europe.CONCEPT.563	At standstill or running operation on 25kV AC network (with four ACMs in operation) the HVAC maximum allowed power consumption shall be 80kVA/car.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.01.Zefiro-Europe.CONCEPT.564	At standstill or running operation, indipendently from catenary (with three ACMs in operation) the HVAC maximum allowed power consumption shall be 45kVA/car. In case of stanstill and three ACMs in operation, under 1,5kV DC network, the the HVAC maximum allowed power consumption shall be 32kVA/car.	Derived Requirement	4S_09.03.05.-.TCMS Software		

	In case of running and three ACMs in operation, under 1,5kV DC network, the the HVAC maximum allowed power consumption shall be 45kVA/car.				
2F_06.01.Zefiro-Europe.CONCEPT.276	At standstill or running operation, indipendently from catenary (with two ACMs in operation) the HVAC shall be ordered to enter reduced power mode and maximum allowed power consumption shall be 32 kVA/car. **Note:** valid also for 1,5kV DC catenary	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.67	At standstill or running operation, indipendently from catenary (with one ACM in operation)indipendently from catenary, HVAC shall be ordered to enter reduced power mode and maximum allowed power consumption shall be 6kVA/car.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.566	When external power supply is connected HVAC maximum allowed power consumption shall be 6kVA/car.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.01.Zefiro-Europe.CONCEPT.535	If 3 ACM:s are not in operation (also during regeneration) only the following loads shall be active. – HVAC in ventilation mode (automatic mode) – 1 Main Air Compressor Note that the Bistro and sockets for seats shall be deactivated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.4.7 System Separation due to Short Circuits or component failure

3.1.4.7.1 Design

If short-circuit is detected by ACM:s they will shutdown. New attempts to restart will be done to try to isolate the smallest possible part of the 3*400V system to isolate. See figure with flowchart for how to locate the short circuit.



3.1.4.7.2 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.235
Description	If short circuit is detected on the main AC bus the separation contactor in this car shall isolate the loads from the bus. The local loads shall be supplied from the local ACM on a two car basis.
Safety level	SIL 0

ID	2F_06.01.Zefiro-Europe.TRS.236
Description	The local load contactor (placed in TT and T cars) shall close if the local AC loads have been separated from the AC bus.
Safety level	SIL 0

ID	2F_06.01.Zefiro-Europe.TRS.124
Description	The 3~ AC bus of the two train halfs, the two 4 car units shall be able to be disconnected with separation contactors. This shall ensure that a short circuit in on 3~ AC system will not affect the whole train.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.448	If **all operational** ACMs indicate 'possible short circuit 3-phase bus' all ACMs shall be ordered to stop.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_06.01.Zefiro-Europe.CONCEPT.466	If a 3 phase bus MCB (8m) has tripped, then contactor 8 on the same 3 phase bus shall be ordered to open and contactor 9 on the same 3 phase bus shall be ordered to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.465	If contactor 8 is open and contactor 9 is closed due to a tripped MCB (8m) then the ACMs shall be ordered to start again.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.504	If **all operational** ACMs indicate 'possible short circuit 3-phase after contactor 9 has been closed due to a tripped MCB (8m) then all ACMs shall be ordered to stop and contactors 6, 7, 8 and 9 of all 3 phase buses shall be ordered to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.464	If the ACMs have been ordered to stop due to indication of 'possible short circuit 3-phase bus' and no MCB (8m) has tripped then contactors 6, 7, 8 and 9 of all 3 phase buses shall be ordered to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.463	When all contactors 6, 7, 8 and 9 have been opened due to 'possible short circuit 3-phase bus', then **all** ACMs shall be ordered to start again.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.462	If a 'possible short circuit 3-phase bus' is detected by any ACM when all contactors 6, 7, 8 and 9 are open, then all ACMs shall be ordered to stop.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.461	If all ACMs have been ordered to stop due to 'possible short circuit 3-phase bus' when contactor 6, 7, 8 and 9 are open, then the contactor 9 on the same 3 phase bus as the ACM that indicated 'possible short circuit 3-phase bus' shall be ordered to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.460	If all ACMs have been ordered to stop due to 'possible short circuit 3-phase bus' when contactor 6, 7, 8 and 9 are open, then the contactors 7 and 8 on the 3 phase buses where **no** 'possible short circuit 3-phase bus' is detected shall be ordered to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.459	If all ACMs have been ordered to stop due to 'possible short circuit 3-phase bus' when contactor 6, 7, 8 and 9 are open, then the contactor 6 shall be ordered to close and all ACM that did not indicate 'possible short circuit 3-phase bus' shall be ordered to start	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.458	If all contactors 6, 7, 8 and 9 have been opened due to 'possible short circuit 3-phase bus', and all ACMS have been ordered to start again but no ACM has indicated 'possible short circuit 3-phase bus', then all contactor 8s shall be ordered to closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.457	If an ACM indicates 'possible short circuit 3-phase bus' when **all** ACMs have been ordered to start again after closing all contactor 8s, then all ACMs shall be ordered to stop.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.456	If all ACMs have been ordered to stop due to an ACM indicating 'possible short circuit 3-phase bus' when all ACMs have been ordered to start again after closing all contactor 8s, then contactor 8 on the same 3 phase bus as the ACM that indicated 'possible short circuit 3-phase bus' shall be ordered to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.455	If a contactor 8 has been ordered to open after an ACM has indicated 'possible short circuit 3-phase bus' then all contactor 7s shall be ordered to close	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.454	If a contactor 8 has been ordered to open after an ACM has indicated 'possible short circuit 3-phase bus' then contactor 6 shall be ordered to close and all ACMs ordered to start again.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_06.01.Zefiro-Europe.CONCEPT.453	If no ACM indicates 'possible short circuit 3-phase bus' when all ACMs have been ordered to start again after closing all contactor 8s, then **all** ACMs shall be ordered to stop and all contactor 7s shall be ordered to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.452	If all contactor 7s have been closed after no detection of 'possible short circuit 3-phase bus' all ACMs shall be ordered to start.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.451	If the operational ACMs in one train half indicate 'possible short circuit 3-phase bus' then the ACMs in that train half that have indicated possible short circuit 3-phase bus' shall be ordered to stop and the contactor 7s shall be ordered to open and the contactor 8s shall be ordered to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.450	When the contactor 7s have been opened and contactor 8s have been closed in that train half that have indicated possible short circuit 3-phase bus' the ACMs shall be ordered to start.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.449	If no ACMs indicate 'possible short circuit 3-phase bus', when all contactor 7s and contactor 8s are closed again after indication of 'possible short circuit 3-phase bus', all ACMs shall be ordered to stop, contactor 6 shall be ordered to close and all ACMs shall be ordered to start	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.505	If **all operational** ACMs indicate 'possible short circuit 3-phase when contactor 6 has closed again after indication of 'possible short circuit 3-phase bus', then the ACMs shall be ordered to stop, both contactor 6 shall be ordered to open and all ACMs shall be ordered to start.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.568	In case of activation of the short circuit sequence the TCMS has to: 1. rise an event (other) stating that the protection sequence has started 2. rise an event (other) reporting the MV configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.01.Zefiro-Europe.CONCEPT.82	Each 4 car unit shall have a bus contactor placed in cars T4 and T5.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.498	Tripped MCB (8M) shall be indicated, high signal when not tripped.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.499	An event shall be set when MCB (8M) has tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.500	Tripped MCB (9M) shall be indicated, high signal when not tripped.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.501	An event shall be set when MCB (9M) has tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.278	The number of short circuit detections of an ACM shall be more than 3 before the ACM is permanently blocked due to short circuit detection. The permanent block shall be reset at master reset.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.551	If fire is detected in the underframe the ACMs in the half of the train with fire shall be stopped and both contactor 6 shall be opened. The half of the train where the fire is detected shall be isolated, i.e. contactor 6 shall not be allowed to close until master reset has been performed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.4.8 Towed with generated power from traction motors

3.1.4.8.1 Function design

- 3AC auxiliary power is generated using braking power from traction motors

- The train is disconnected from the catenary.
- Traction chain backwards from traction motor to DC-link has enough functionality to generate power to the 3AC network.
- All 4 Auxiliary converters have full functionality

3.1.4.8.2 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.244
Description	The Towing train shall accelerate to at least 50km/h or more to enable three phase loads on the bus in a Towed train.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.245
Description	The three phase loads in the Towed train shall be disconnected during towed when train speed is lower than 45 km/h.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.246
Description	In a Towed train the battery charging and essential braking, safety and comfort functions shall be maintained if possible.
Safety level	N/A

The train is set to "Towed mode" if the Towing switch is turned on the driver's desk. See CTC SFRS and Brake SFRS for more information.
To maintain battery charging and essential braking, safety and comfort functions during regeneration the battery charging will be activated when 3AC network is activated and 1 main compressor and 1 HVAC per car will be allowed.

To fulfil ACM function during the entire speed range the 3 phase load needs to be limited.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.555	If the train is set to "Towed mode", the "train half separation contactor" (6) shall be ordered to close. Separation contactor 6 shall be ordered to open when the train is no longer set to "Towed mode" AND train is at standstill.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.527	If “Towed mode” is active and the train speed >50km/h then regeneration during towed is considered active and the loads shall be allowed to start.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.529	When at least 2 ACMs are in operation during regeneration the following loads shall be allowed: – 1 Main compressor – HVAC shall be ordered to enter reduced power mode, maximum allowed power consumption 20kVA/car	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.528	If regeneration during towed is active and the speed drops to <45km/h the loads shall be stopped according to the following sequence: • At speed < 45 km/h: Disable HVAC loads • At speed < 45 km/h: Disable main compressors	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

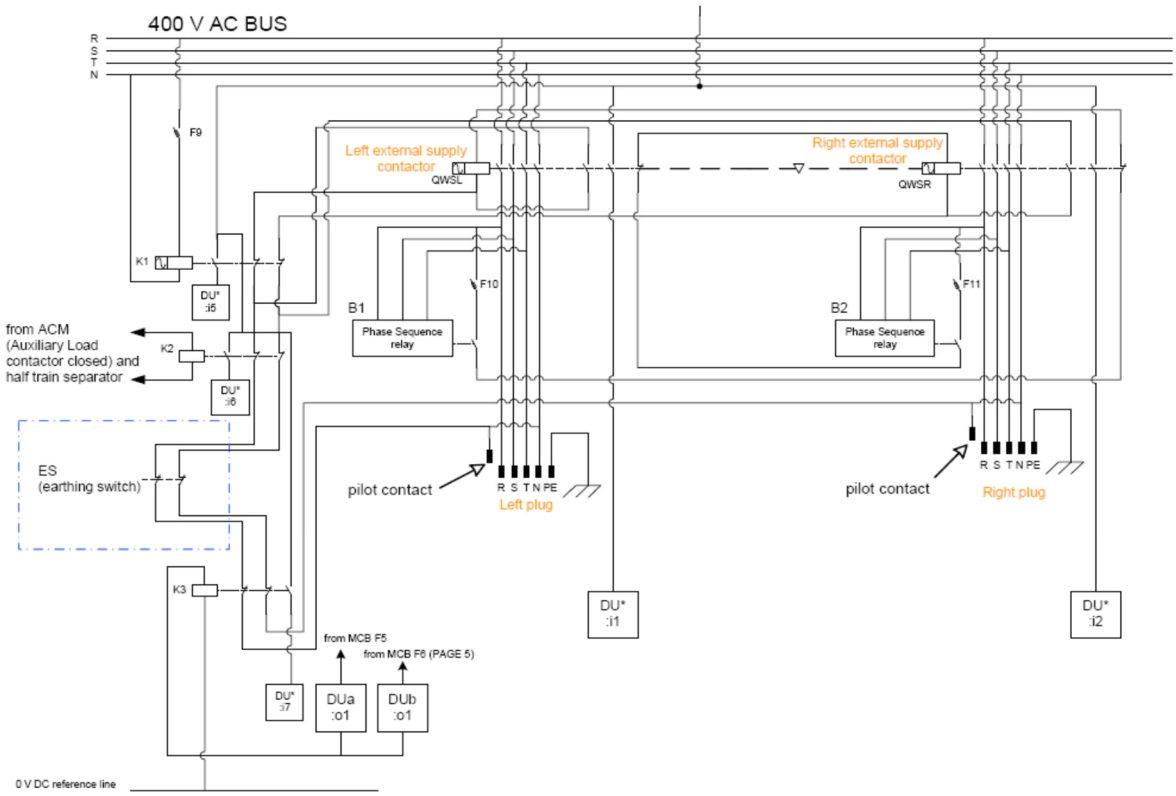
3.1.5 External Power Supply

3.1.5.1 Function Design

General Conditions for workshop supply:
- The train is disconnected from the catenary.
- External power supply plug is connected and activated.
- Workshop power supply is limited to 125 A per plug
- Auxiliary and traction converters are deactivated and disconnected.

- Only one plug can be connected per each 4-car train unit.
- Target is to provide enough power to enable operational functions in conjunction with normal workshop operation.
- Phase sequency check will be done before the contactor closes.

The limitation of 125 A per 4-car train, provided by external power supply determines which actions can be taken inside the workshop. The first connection with external power supply activates the interlocking and inhibits multiple connections.



3.1.5.2 Normal condition

Two workshop inlets are placed on each train half. Only one inlet is allowed to use to supply the train half.

3.1.5.3 Failure cases

1. Contactors not possible to close
2. Contactors not possible to close
3. Contactors not possible to close

3.1.5.4 Failure detection

1. Battery charger is not active
2. Battery charger is not active
3. Battery charger is not active

3.1.5.5 Failure handling

1. Change the wiring on the external cable, due to wrong phase sequency
2. Check the fuses in the workshop, measure the voltage on the cable
3. Check the MCB:s in distribution box

3.1.5.6 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.132
Description	Plugs of type CEE 3AC 400V 125 A according to EN 60309-1/2 with poles of type 3P+N+PE as well as pilot contacts shall be used and mounted in the T cars.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.156	Plugs of type CEE 3AC 400V 125A with poles 3P+N+PE and pilot contacts shall be used.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_06.01.Zefiro-Europe.TRS.138
Description	A separation contactor isolating the on board 3~ bus from the 3~ inlets in the T cars shall be provided.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.129
Description	A 3~ AC load or separation contactor shall be provided in the cars where external inlets are located.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.158	4 pole separation contactor shall be used for the external inlets.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_06.01.Zefiro-Europe.TRS.137
Description	The separation contactor shall close to supply the 3~ AC bus from the external inlet if no voltage is detected on the 3~ 400 V AC bus.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.143
Description	External supply inlets shall be interlocked so that only one can feed the 3AC train bus bar.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.160	Contactors for external supply shall not be able to close if no cable is connected to the inlet.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.328	Contactors for external supply shall not be able to close if voltage is detected on the 3AC bus.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.341	Contactors for external supply shall not be able to close if no voltage is detected on the 3AC inlet (cable is not powered).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.317	Give clearance (remove open order) for the "external supply contactor" when following conditions are fulfilled: – No ACM activated – No pantograph is raised – At least one of the contactor 6 is open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.345	The "external supply contactor" shall close when all of the following conditions are fullfilled: – Phase sequency ok – No power is detected on the 3AC Bus 1 – No "external supply contactor" in the train half is closed – Contactors are in the positions stated for half train normal mode – No earthing switch closed	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

	– No 'External AC disconnection' from TCMS (order "open external supply contactor" is not set)				
2F_06.01.Zefiro-Europe.CONCEPT.502	Separation contactor 6 shall be ordered to open if any of the following conditions applies – 30s after the pantographs are lowered AND train is at standstill AND train is not in towed mode – At train deactivation (battery contactors about to be opened) – 3-phase external power supply is connected	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_06.01.Zefiro-Europe.TRS.136
Description	The separation contactor shall open to isolate the external inlet from the ACM if any ACM is active and any ACM load contactor closed. The inlet shall be disconnected from the 3~ bus as long as voltage is detected. It shall be ensured that 3 phase voltage is never active on a 3 phase inlet-plug while not connected to an external plug.
Safety level	SIL 2

ID	2F_06.01.Zefiro-Europe.TRS.147
Description	In operation mode external power supply shall be mutually locked.
Safety level	N/A

To ensure that the external supply inlets are powerless when not connected there is an interlock between the auxiliary load contactors at the ACM and the external supply contactor.
Moreover, the external supply contactor can be closed only if external power supply plug is connected and activated.

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.162	The "external supply contactor" shall be blocked from closing as long as any ACM:s load contactor is closed.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_06.01.Zefiro-Europe.CONCEPT.539	The status of "ACM load contactor" shall be available for conventional train control.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_06.01.Zefiro-Europe.CONCEPT.443	Position of the external supply contactors (left and right) shall be monitored.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_06.01.Zefiro-Europe.CONCEPT.442	Status of the 'external supply contactors left and right', load contactors (6), (7), (8), (9), MCB (8m), MCB (9m) and auxiliary load contactor at the ACM shall be indicated on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

ID	2F_06.01.Zefiro-Europe.TRS.131
Description	The connection of an external power supply plug shall be recognised by the train and the traction shall be blocked as long as an external power supply is applied.
Safety level	N/A

ID	2F_06.01.Zefiro-Europe.TRS.144
Description	Traction shall be locked when at least one external supply cable is connected to the train (supply voltage is detected).
Safety level	N/A

Traction with a cable connected to the train is not allowed. It is a danger to the workshop personnel and to the electrical installation in the workshop. This interlock can be a reliability issue if the detection device wrongly detects a cable. Therefore the driver shall have the possibility to override this interlock, after checking that no external power supply is plugged and deducing that it may be a fault in detection circuit.

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.164	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_06.01.Zefiro-Europe.CONCEPT.205	If an external power supply is detected, but no power is detected on the 3AC Bus 1, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.554	The IDU shall provide an isolation command of each external plug. when this command is activated by the driver, the TCMS shall constantly energize the output to isolate that plug. The ordered isolation command will be automatically reset at the LCB closure.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_06.01.Zefiro-Europe.CONCEPT.559	In case any plug is detected inserted (based on the status of relay QWSL or QWSR) at speed > 3km/h or with the HV enabled the affected external plug shall be constantly disabled by TCMS. The automatically disabled plug will be re-enabled as soon as all panto will be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.01.Zefiro-Europe.CONCEPT.560	In case any plug is detected inserted (based on the status of relay QWSL or QWSR) at speed > 3km/h or with the HV enabled an event for the maintenance shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

ID	2F_06.01.Zefiro-Europe.TRS.145
Description	External supply shall start the battery charging.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.176	At least 1 battery charger shall be directly (no controlled contactors) connected to the 3AC bus.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_06.01.Zefiro-Europe.TRS.146
Description	A power management shall take care of the ratio between available power from external power supply and the load of the internal consumers.
Safety level	N/A

The loads that are not controlled by train control will always be powered when the 3AC Bus is powered regardless of the source of power (internal or external).

ID	2F_06.01.Zefiro-Europe.TRS.148
Description	Phase rotation protection shall protect the auxiliary motors against operation in wrong direction
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.87	The external supply contactors shall be blocked from closing if the phase rotation of the external power differs from the phase rotation expected by the auxliliary system in the train.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		N/A

3.1.6 Earthing switch

3.1.6.1 Function Design

3.1.6.2 Normal condition

For maintenance on the 3-phase bus the train is equipped with earthing switches on each train half. These are located where the workshop inlets are placed.

3.1.6.3 Failure cases

1. Wrong position indication from earthing switch, indicating closed when in reality the earthing switch is open
2. Wrong position indication from earthing switch, indicating open when in reality the earthing switch is closed.

3.1.6.4 Failure detection

1. If closed indication the TCMS will shut down the train.
2. If open the earthing switch gives open indication and in reality is closed and the train are activated, the overload of the ACMs are supervised by the ACM itself. External supply outlets are protected in the workshop.

3.1.6.5 Failure handling

1. Repaired during corrective maintenance.
2. Repaired during corrective maintenance.

3.1.6.6 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.152
Description	Manual erthing the 3AC-bus shall be provided for maintenance staff during maintenance procedures. It shall connect the active conductiors to earth.
Safety level	N/A

Earthing of the auxiliary power system is a part of the so called key chain. To be able to open certain boxes and cabinets keys must be used that can only be obtained by doing the proper isolation, in this case earthing.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.227	If the earthing switch is in earthed position no ACM shall be allowed to start	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.01.Zefiro-Europe.CONCEPT.344	If the earthing switch is in earthed position no external supply contactors shall be allowed to be closed	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.486	The status of earthing switch shall be connected to TCMS DI	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.HR08	If the earthing switch is in earthed position no external supply contactors shall be allowed to be closed from TCMS	Derived Requirement	4S_09.02.09.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.441	If the earthing switch is in earthed position an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.1.7 230V, 50Hz Sockets

3.1.7.1 Function Design

Each seat is equipped with socket with power 230V, 50Hz

3.1.7.2 Normal condition

Each seat are supplied with power 230V, 50Hz

3.1.7.3 Failure cases

1. No power supply to the socket
2. No power supply to the socket

3.1.7.4 Failure detection

1. Miniature circuit breaker tripped, an event will be set
2. RCD tripped, an event will be set

3.1.7.5 Failure handling

1. Repaired in maintenance
2. Repaired in maintenance

3.1.7.6 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.100
Description	For each passenger seat a 1-phase 230 V 50 Hz supply socket of Italian type shall be installed. Additional 230 V 50 Hz sockets shall be foreseen for maintenance and cleaning purposes.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.102	Each passenger seat shall have 1-phase 230V supply socket	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.200	Two power sockets per car for cleaning and maintenance purpose shall be installed.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_06.01.Zefiro-Europe.TRS.243
Description	All 1-phase 230V 50 Hz supply sockets shall be disconnected in case of fire in the passenger compartment.
Safety level	

As the train is fitted with fire extinguisher there is a risk for electric shock if the 1 phase 230V 50Hz supply sockets are powered when water is sprayed in the passenger compartment.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.01.Zefiro-Europe.CONCEPT.331	If fire is detected in the passenger compartment the 1 phase 230V 50Hz supply sockets, both for passenger and maintenance, shall be disconnected. (TCMS DO false at fire to disconnect the supply)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.536	If fire is detected in the passenger compartment the 1 phase 230V 50Hz supply sockets, both for passenger and maintenance, shall be disconnected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.439	If fire is detected in the passenger compartment the 1 phase 230V 50Hz supply Bistro and Toilets shall be disconnected. (TCMS DO false at fire to disconnect the supply)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.01.Zefiro-Europe.CONCEPT.537	If fire is detected in the passenger compartment the 1 phase 230V 50Hz supply Bistro and Toilets shall be disconnected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.8 Mini Circuit Breaker Supervision

3.1.8.1 Function Design

3.1.8.2 Normal condition

Miniature circuit breakers are supervised, an event indicating which cabinet has a tripped MCB

3.1.8.3 Failure cases

1. Miniature circuit breaker tripped

3.1.8.4 Failure detection

1. An event will be set on IDU indicating in which cabinet the tripped MCB is located

3.1.8.5 Failure handling

1. Repaired in maintenance

3.1.8.6 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.128
Description	All 3~ AC MCB shall be supervised. A tripped MCB shall be indicated by a diagnostic message indicating the location of the tripped MCB.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.274	Non permanent connected consumers shall be protected with RCDs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.193	Tripped MCB shall be indicated using one event per cabinet or box.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.434	Tripped RCD shall be indicated using one event per cabinet or box.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.01.Zefiro-Europe.CONCEPT.435	The status switch for all MCBs in a cabinet or box shall be daisy chained to allow supervision of the MCB status on cabinet or box level and shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.01.Zefiro-Europe.CONCEPT.436	The status switch for all RCDs in a cabinet or box shall be daisy chained to allow supervision of the RCD status on cabinet or box level and shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.9 Events

3.1.9.1 Function Design

The ACMs have event handling system included (a process client reporting events directly into the CDU). The events will be transferred directly to the ODBS on the CDU for storing and indication.

3.1.9.2 Requirements

ID	2F_06.01.Zefiro-Europe.TRS.238
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_06.01.Zefiro-Europe.CONCEPT.261	Diagnostics from the ACM shall be reported via an IP-process client directly into the train diagnostic system.	Derived Requirement	4S_05.-.Propulsion	EVENT	N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_06.01.Zefiro-Europe.CONCEPT.28	None		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_06.01.Zefiro-Europe.CONCEPT.29	None		

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_06.01.Zefiro-Europe.CONCEPT.335	Train half separation contactor closed. (contactor 6)		2F_06.01.Zefiro-Europe.CONCEPT.478
2F_06.01.Zefiro-Europe.CONCEPT.336	Bus 1 contactor closed. (contactor 7)		2F_06.01.Zefiro-Europe.CONCEPT.472
2F_06.01.Zefiro-Europe.CONCEPT.337	Bus 2 contactor closed. (contactor 8)		2F_06.01.Zefiro-Europe.CONCEPT.469
2F_06.01.Zefiro-Europe.CONCEPT.338	Bus 1 and 2 separation contactor closed. (contactor 9)		2F_06.01.Zefiro-Europe.CONCEPT.475
2F_06.01.Zefiro-Europe.CONCEPT.340	Bus 2 miniature circuit breaker not tripped. (8M)		2F_06.01.Zefiro-Europe.CONCEPT.499
2F_06.01.Zefiro-Europe.CONCEPT.339	Bus 1 and 2 separation miniature circuit breaker not tripped. (9M)		2F_06.01.Zefiro-Europe.CONCEPT.501
2F_06.01.Zefiro-Europe.CONCEPT.516	MCB not tripped.		2F_06.01.Zefiro-Europe.CONCEPT.193
2F_06.01.Zefiro-Europe.CONCEPT.517	RCD not tripped.		2F_06.01.Zefiro-Europe.CONCEPT.434
2F_06.01.Zefiro-Europe.CONCEPT.513	External supply contactor left closed		2F_06.01.Zefiro-Europe.CONCEPT.442
2F_06.01.Zefiro-Europe.CONCEPT.541	External supply contactor right closed		2F_06.01.Zefiro-Europe.CONCEPT.442
2F_06.01.Zefiro-Europe.CONCEPT.514	Earthing switch closed position.		2F_06.01.Zefiro-Europe.CONCEPT.441
2F_06.01.Zefiro-Europe.CONCEPT.515	Earthing switch open position.		2F_06.01.Zefiro-Europe.CONCEPT.441

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_06.01.Zefiro-Europe.CONCEPT.31	Close train half separation contactor. (contactor 6)		2F_06.01.Zefiro-Europe.CONCEPT.323 2F_06.01.Zefiro-Europe.CONCEPT.476 2F_06.01.Zefiro-Europe.CONCEPT.477 2F_06.01.Zefiro-Europe.CONCEPT.478

2F_06.01.Zefiro-Europe.CONCEPT.332	Close Bus 1 contactor. (contactor 7)		2F_06.01.Zefiro-Europe.CONCEPT.323 2F_06.01.Zefiro-Europe.CONCEPT.470 2F_06.01.Zefiro-Europe.CONCEPT.471 2F_06.01.Zefiro-Europe.CONCEPT.472
2F_06.01.Zefiro-Europe.CONCEPT.333	Close Bus 2 contactor. (contactor 8)		2F_06.01.Zefiro-Europe.CONCEPT.323 2F_06.01.Zefiro-Europe.CONCEPT.467 2F_06.01.Zefiro-Europe.CONCEPT.468 2F_06.01.Zefiro-Europe.CONCEPT.469
2F_06.01.Zefiro-Europe.CONCEPT.334	Close Bus 1 and 2 separation contactor. (contactor 9)		2F_06.01.Zefiro-Europe.CONCEPT.323 2F_06.01.Zefiro-Europe.CONCEPT.473 2F_06.01.Zefiro-Europe.CONCEPT.474 2F_06.01.Zefiro-Europe.CONCEPT.475
2F_06.01.Zefiro-Europe.CONCEPT.343	Open external supply contactor		2F_06.01.Zefiro-Europe.CONCEPT.317
2F_06.01.Zefiro-Europe.CONCEPT.542	Close 230V sockets in passenger compartment		2F_06.01.Zefiro-Europe.CONCEPT.536
2F_06.01.Zefiro-Europe.CONCEPT.543	Close 230V supply to Bistro and Toilets		2F_06.01.Zefiro-Europe.CONCEPT.537

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_06.01.Zefiro-Europe.CONCEPT.490	ACM bus active		2F_06.01.Zefiro-Europe.CONCEPT.445 2F_06.01.Zefiro-Europe.CONCEPT.446
2F_06.01.Zefiro-Europe.CONCEPT.506	ACM is faulty		2F_06.01.Zefiro-Europe.CONCEPT.447
2F_06.01.Zefiro-Europe.CONCEPT.507	ACM is cut out		2F_06.01.Zefiro-Europe.CONCEPT.447
2F_06.01.Zefiro-Europe.CONCEPT.508	Status: DC-link charged		2F_06.01.Zefiro-Europe.CONCEPT.447
2F_06.01.Zefiro-Europe.CONCEPT.511	Status: ACM is in operation		2F_06.01.Zefiro-Europe.CONCEPT.67 2F_06.01.Zefiro-Europe.CONCEPT.276 2F_06.01.Zefiro-Europe.CONCEPT.445 2F_06.01.Zefiro-Europe.CONCEPT.446
2F_06.01.Zefiro-Europe.CONCEPT.512	Possible short circuit 3-phase bus		2F_06.01.Zefiro-Europe.CONCEPT.448 2F_06.01.Zefiro-Europe.CONCEPT.451 2F_06.01.Zefiro-Europe.CONCEPT.453 2F_06.01.Zefiro-Europe.CONCEPT.457 2F_06.01.Zefiro-Europe.CONCEPT.462 2F_06.01.Zefiro-Europe.CONCEPT.504 2F_06.01.Zefiro-Europe.CONCEPT.505

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_06.01.Zefiro-Europe.CONCEPT.33	Disable start of auxiliary converter		2F_06.01.Zefiro-Europe.CONCEPT.445
2F_06.01.Zefiro-Europe.CONCEPT.496	Disable idle start of auxiliary converter		2F_06.01.Zefiro-Europe.CONCEPT.445
2F_06.01.Zefiro-Europe.CONCEPT.32	Status: 3-phase voltage bus active		2F_06.01.Zefiro-Europe.CONCEPT.246
2F_06.01.Zefiro-Europe.CONCEPT.491	Reset of ACM		2F_06.01.Zefiro-Europe.CONCEPT.315
2F_06.01.Zefiro-Europe.CONCEPT.HR05	Status 3-phase voltage bus active main transformer		2F_06.01.Zefiro-Europe.CONCEPT.HR01
2F_06.01.Zefiro-Europe.CONCEPT.HR06	Status 3-phase voltage bus active converter cooling		2F_06.01.Zefiro-Europe.CONCEPT.HR02